

Nutrition in Cholestatic Liver Disease.

Picture an infant of three months who had a Kasai operation for biliary atresia six weeks ago. The jaundice has not fully cleared. The parents tell you she feeds well, and the scale in clinic shows that her weight is creeping up. Her abdomen is full, her liver and spleen are large, and her arms look thin. The question for this chapter is simple to ask and harder to answer. Is this child well nourished, and if not, what exactly should we feed her, how much, by which route, and how will we know it is working?

The main idea of the chapter is this. In a child with cholestasis, malnutrition is not a side effect that we can deal with later. It is part of the disease. It comes from at least four separate mechanisms that act at the same time, it hides behind a reassuring weight, and it predicts whether the child keeps the native liver, how the child fares on the transplant list, and how well the child recovers after a transplant. So nutrition needs the same precision that we give to bilirubin or the international normalised ratio, usually shortened to INR.

A short note on sources. The most important document is the joint position paper of the North American and European societies for paediatric gastroenterology, hepatology and nutrition, NASPGHAN and ESPGHAN, on nutrition support of children with chronic liver disease. It was written by Mouzaki and colleagues and published in the Journal of Pediatric Gastroenterology and Nutrition in 2019. For children awaiting transplant, the American liver society, AASLD, published an evaluation guideline with Squires as first author in 2014, and that guideline has now been replaced by a 2026 guideline from AASLD, the American Society of Transplantation and NASPGHAN. The European liver society, EASL, published a nutrition guideline in 2019, but it covers adults with cirrhosis only, and we will use it as a contrast. Much of the evidence on vitamins comes from the North American research network on childhood liver disease, called ChiLDRen, and much of the transplant evidence comes from the registry called Studies of Pediatric Liver Transplantation, or SPLiT.

A few words first. Cholestasis means reduced flow of bile. Bile acids are the detergents in bile. In the gut they gather into tiny packages called micelles, and micelles are what carry dietary fat and the fat-soluble vitamins A, D, E and K to the wall of the intestine. Long-chain triglycerides, which we will call LCT, are the ordinary fats of milk and food. Medium-chain triglycerides, which we will call MCT, are shorter fats with eight to ten carbon atoms. Essential fatty acids are the fats the body cannot make for itself, mainly linoleic acid and alpha-linolenic acid. A z-score tells you how many standard deviations a measurement is above or below the average for age and sex. A z-score of minus two is roughly the second or third centile. Mid-upper arm circumference, which we will call MUAC, is the circumference of the upper arm halfway between the shoulder and the elbow. The triceps skinfold is the thickness of a pinch of skin and fat over the back of the upper arm, measured with calipers.

Sarcopenia means loss of muscle mass and strength.

Part one. Why the cholestatic infant starves.

There are four mechanisms, and it helps to take them one at a time.

The first is that bile does not arrive. When bile flow is reduced, too few bile acids reach the duodenum to form micelles. Without micelles, long-chain fat cannot be carried to the gut wall, and neither can vitamins A, D, E and K. So the child loses fat calories in the stool and slowly becomes deficient in all four vitamins. A useful warning is that this fat loss can be silent. The stools of a cholestatic infant are pale, and they may be formed rather than greasy, so you cannot rely on the parents noticing steatorrhoea.

The second mechanism is that the sick liver burns more energy. This has been measured. In 1989, Pierro and colleagues measured energy expenditure in infants with biliary atresia and found it was sixty-eight kilocalories per kilogram per day. That is twenty-nine percent higher than normal, and the difference was highly significant. Even more striking, only about one third of the energy the infants could absorb and use, thirty-seven kilocalories per kilogram per day, was stored as new tissue. In 2003, Greer and colleagues studied twenty-one children under two years of age with end-stage liver disease and compared them with fifteen controls. They used a four-compartment model of body composition together with indirect calorimetry, which measures oxygen use and carbon dioxide output. Resting energy expenditure per kilogram was twenty-seven percent higher in the children with liver disease. Their respiratory quotient was lower, which means they were burning relatively more fat. And because the authors measured body cell mass directly, thirty-three percent of body weight in the sick children against twenty-nine percent in controls, they could show that this was a true hypermetabolic state, not an artefact of an odd body composition. The AASLD guideline of 2014 summarises the practical point: children with chronic liver disease need twenty to eighty percent more calories than healthy children to grow.

The third mechanism is that intake falls. These children are often anorexic and nauseated, and they may vomit. A large liver and spleen and any ascites take up space in the abdomen, so the stomach fills early. Pruritus disturbs sleep and makes feeding a battle. So at the very moment when the child needs more energy, the child takes in less.

The fourth mechanism is that fuel is mishandled. The cholestatic liver stores little glycogen, so the child tolerates fasting poorly. The 2019 position paper describes how amino acids are diverted to make glucose, and how this leads to increased oxidation of the branched-chain amino acids, which are the amino acids that muscle relies on. In other words, the child burns muscle protein for fuel.

Put these together and you see why feeding a normal amount of normal formula cannot work. The deficit is malabsorption plus hypermetabolism plus low intake plus a shift towards burning muscle. The answer has to address all four: more energy, the right kind of fat, protein that is not restricted, and a route of feeding that bypasses a poor appetite.

Part two. Long chains need bile, medium chains do not.

This is the single most useful piece of physiology in cholestatic nutrition, so it deserves a careful walk-through.

Long-chain fats, with sixteen to twenty carbon atoms, are poorly soluble in water. To be absorbed they must first be broken down by pancreatic lipase and then carried in micelles to the gut wall. Inside the gut cell they are rebuilt into triglycerides, packed into particles called chylomicrons, and sent into the lymph. When bile acids are scarce, the very first step, the micelle, fails.

Medium-chain fats, with eight to ten carbon atoms, behave differently. They are soluble enough in water to be absorbed without micelles. They pass directly into the portal vein and travel to the liver. So they deliver calories even when bile is absent. They give eight point three kilocalories per gram, a little less than the nine kilocalories per gram of long-chain fat.

There is a catch, and it is important. MCT oils contain no essential fatty acids at all. Linoleic acid and alpha-linolenic acid are long-chain fats. They can only come from LCT. So MCT is best thought of as a fuel rather than a complete fat. If you push the proportion of MCT too high, you trade steatorrhoea for essential fatty acid deficiency.

The 2019 position paper turns this into numbers. Start with about thirty percent of fat calories as MCT and seventy percent as LCT. Never let MCT exceed eighty percent of fat, and go lower in severe cholestasis. And make sure that at least three percent of total energy comes from linoleic acid, and point seven to one percent from alpha-linolenic acid.

Part three. Weight lies in the cholestatic child.

Now return to the infant in the opening scene, with a weight that is creeping up. Why should we not trust it?

The reason is simple. A large liver, a large spleen, ascites and oedema all add weight that is not tissue. So the scale can show steady gain while fat and muscle are being lost. The arm has no organs and usually little oedema, so measurements of the arm reflect real stores of fat and muscle. As long ago as 1990, Sokol and Stall showed that arm circumference and triceps skinfold were the most sensitive measures of nutritional status in children with chronic liver disease.

A Brazilian study by Silva and colleagues, published in 2016, shows the size of the gap. They studied ninety-one children with cholestasis, with a median age of twelve months. If you used weight-for-age below minus two standard deviations, thirty-three percent were malnourished. If you used height-for-age, thirty point eight percent. But mid-upper arm circumference showed depletion in forty-three point eight percent of the eighty-nine children measured, and triceps skinfold showed depletion in sixty-four percent. In other words, the arm found about twice as much malnutrition as the scale.

The 2019 position paper therefore recommends serial measurement of mid-upper arm circumference and triceps skinfold, every two weeks to every three months depending on the child. The AASLD guideline of 2014 says the same thing in different words: these arm measurements are the most reliable anthropometric assessments, and reliance on weight alone may overestimate nutritional adequacy.

It helps to know what each measurement can and cannot tell you. Weight is total mass, including fluid and organs, so use it only as a trend. Length or height is a slow, cumulative marker. It falls late, but once it falls it tells you the deficit has been going on for a long time, and it matters a great deal for transplant allocation, as we will see. Mid-upper arm circumference reflects the muscle and fat of the arm. It needs a trained person and a tape placed exactly at the midpoint. Triceps skinfold reflects fat stores. It has some error between observers and can be distorted by oedema in very sick children. Head circumference matters under two years of age, but it is spared until late. Finally, muscle mass can be measured with computed tomography of the psoas muscle, or with dual-energy X-ray absorptiometry, called DEXA. These are not screening tools, but they are useful in a transplant work-up, especially if a scan has been done for another reason.

Part four. Muscle is lost before weight.

How common is malnutrition in these children? The 2019 position paper cites several cohorts. In one, thirty-nine percent of thirty-eight infants with neonatal cholestasis were malnourished by mid-upper arm circumference. In another, sixty-nine percent of fifty-one children listed for liver transplant, with a mean age of three point seven years, had impaired growth.

Muscle has been studied directly. In 2018, Lurz and colleagues measured the area of the psoas muscle at the level of the third and fourth lumbar vertebrae on computed tomography, corrected for height. They compared twenty-three infants and young children with end-stage liver disease with forty-six matched controls. The psoas area was significantly lower in the children with liver disease. In 2019, Mager and colleagues used DEXA in forty-one children after liver transplant and found that forty percent were sarcopenic. Sarcopenia that persisted after transplant was linked to poorer growth and to more readmissions to hospital.

The lesson is that muscle does not come back on the day of transplant. If a child is heading for transplant, the muscle you build before the operation is the muscle the child will recover with.

Part five. Growth after the Kasai operation predicts the transplant.

Is growth failure just a marker of sick liver, or is it an independent sign that we should act on? The best paediatric data come from a multicentre North American study by DeRusso and colleagues, published in *Hepatology* in 2007.

They followed one hundred infants after the Kasai operation, which is also called hepatoportoenterostomy, at nine centres between 1997 and 2000. Forty-six had a poor outcome by the age of twenty-four months. That means forty-two had a liver transplant and four died. The study chart compared weight-for-age z-scores in the infants who kept their own liver with those who had a poor outcome.

Six months after the Kasai operation, the average weight z-score was minus one point two in infants who kept their liver and minus two point one in infants with a poor outcome. That difference was highly significant. The authors then looked at a group of infants with a moderate total bilirubin, between two and six milligrams per decilitre, three months after surgery. Even within that group, the weight z-score was minus one point zero in those who did well and minus two point zero in those who went on to transplant or death. Growth of both weight and length was slower in the poor outcome group.

So a falling weight z-score after the Kasai operation is a prognostic sign, not just a dietetic problem. It should prompt early intensification of nutrition and early discussion with a transplant centre.

Part six. Growth failure is in the score.

Growth also matters on the transplant waiting list. Utterson and colleagues analysed the SPLIT registry in 2005. They included seven hundred and fifty-five children with biliary atresia who were listed for transplant, of whom five hundred and sixty-seven had a transplant. The three-year survival after transplant was eighty-eight percent, and infection caused thirty-seven percent of the deaths after transplant. Growth failure, defined as a height or weight z-score below minus two, was a risk factor for death and graft loss. The authors concluded that nutritional status should prompt aggressive nutritional support.

The paediatric end-stage liver disease score, called PELD, is used in the United States to rank children on the waiting list. In the original score, growth failure was a simple yes or no. If the lower of the height or weight z-score was below minus two, the score added a fixed amount, zero point six six seven in the score's equation. About thirty-one point four percent of listed children met that definition. In 2021, Hsu and colleagues refitted the score using data from four thousand eight hundred and seventy-six children. They replaced the yes or no rule with a

continuous term, so that each step of growth failure counts, down to a z-score of minus five. The ability of the score to tell apart children who would die on the list from those who would not, measured as the area under the curve, rose from zero point seven nine nine to zero point eight five four.

The practical message is that a child with a z-score of minus two point one and a child with a z-score of minus four were once treated the same. Now they are not. So length and weight must be measured carefully, because small differences now move the child on the list.

Part seven. How much to feed, and of what.

We can now set targets, using table three of the 2019 position paper for cholestasis.

Energy should be about one hundred and thirty percent of the requirement for age, and at least that much in end-stage disease. If indirect calorimetry is available, use it. In practice, this means concentrating the formula and adjusting the target according to arm measurements and length, not weight.

Protein should be one hundred and thirty to one hundred and fifty percent of the requirement for age. The AASLD guideline of 2014 expresses the same idea as two to four grams per kilogram per day. Protein should not be restricted. The only exception in the position paper is severe encephalopathy, and even then restriction should last no more than two to three days.

Fat should provide thirty to fifty percent of energy. As we said, start with MCT at thirty percent of fat and LCT at seventy percent, and adjust. Linoleic acid should provide at least three percent of energy and alpha-linolenic acid point seven to one percent.

Carbohydrate should provide forty to sixty percent of energy. Glucose polymer modules can add calories without adding volume, and long fasts should be avoided because glycogen stores are small.

Then there is the route. The position paper recommends nasogastric or nasojejunal tube feeding if energy goals are not met for more than two weeks, or if the anthropometric trend is poor over two to four weeks. Parenteral nutrition is used when enteral feeding is not tolerated or fails to achieve growth. A practical tip is that continuous feeds overnight leave the child hungry during the day, which protects oral feeding skills and daytime appetite.

What about branched-chain amino acids? The position paper mentions a single small randomised trial, by Chin and colleagues in 1992, in which a formula enriched in branched-chain amino acids led to better anthropometric outcomes in children with end-stage liver disease. It is a reasonable option, but the evidence is thin.

Part eight. More MCT is not better.

It is tempting to think that if MCT works around the bile defect, more MCT must be better. It is not.

With too little MCT, fat calories are lost in the stool, weight and skinfold fall, and the fat-soluble vitamins are lost along with the fat. With too much MCT, the child runs out of essential fatty acids. In 1991, Pettei and colleagues described four infants with liver disease fed a formula very high in MCT. All four had biochemical essential fatty acid deficiency. Their arachidonic acid was low, and an abnormal fatty acid called Mead acid had appeared, which the body makes when it lacks linoleic acid. Linoleic acid itself was low in three of the four. Clinically, essential fatty acid deficiency can show as a scaly rash, poor growth, low platelets and infections.

How should you test for it? The 2019 position paper recommends measuring linoleic acid, alpha-linolenic acid, eicosapentaenoic acid and docosahexaenoic acid, the last two usually shortened to EPA and DHA, against age-specific reference ranges. It states plainly that a ratio of triene to tetraene above zero point two, the classic test, is not enough on its own. Test children who have symptoms or whose fat-soluble vitamin deficiency will not correct. Essential fatty acids and fat-soluble vitamins should be checked every three to six months, and monthly in severely cholestatic infants.

How should you fix it? Add an LCT oil so that linoleic acid is at least three percent of energy, and consider a source of DHA and arachidonic acid. Socha and colleagues showed in 1997 that essential fatty acid status is related to serum bilirubin. The sicker the liver, the harder you have to work to get long-chain fat absorbed. So there is no single formula for cholestasis. The right share of MCT is the lowest one that controls steatorrhoea.

Part nine. The fat-soluble vitamins.

Each of the four vitamins has a starting dose, a laboratory marker and a trap.

Vitamin A. The position paper recommends five thousand international units a day for a child under ten kilograms and ten thousand international units a day for a child over ten kilograms. Deficiency is best judged by the molar ratio of retinol to retinol-binding protein. A ratio below zero point eight suggests deficiency. If serum retinol is below twenty micrograms per decilitre, a modified relative dose response test can confirm it. The trap is toxicity. The window between deficiency and toxicity is narrow, and too much vitamin A can itself cause liver fibrosis. So never dose vitamin A without checking levels.

Vitamin D. The position paper recommends cholecalciferol, two thousand to five thousand international units a day. The ChiLDReN studies defined deficiency as a twenty-five hydroxy vitamin D level below fifteen nanograms per millilitre. One caution: the target printed in the 2019 paper appears to have a unit error, so check the units before you quote it. The adult EASL

guideline suggests supplementing below twenty nanograms per millilitre and aiming above thirty. Watch for rickets and fractures, and check calcium and alkaline phosphatase.

Vitamin E. The preferred form is a water-soluble preparation called TPGS, which stands for d-alpha-tocopheryl polyethylene glycol succinate, at fifteen to twenty-five international units per kilogram per day. The trap here is the serum level. Cholestatic children often have high blood lipids, and vitamin E travels with lipids, so a deficient child can have a normal serum vitamin E. The correct test is the ratio of vitamin E to total lipids. A ratio below zero point six milligrams per gram is deficient in children aged one to twelve years, and below zero point eight in older children. Deficiency causes neuropathy, ataxia and haemolysis.

Vitamin K. The dose is two to five milligrams a day by mouth, and one to ten milligrams intravenously may be needed. The INR is the usual marker. A protein called PIVKA-II, which stands for protein induced by vitamin K absence, is more sensitive but not widely available. The trap is that a normal INR does not prove adequate vitamin K. The position paper notes that deficiency relevant to bone can occur in thirty-nine percent of children despite a normal INR. A normal INR proves the clotting factors are being made. It does not prove the child has enough vitamin K for bone.

Part ten. Supplemented and still deficient.

The ChiLDReN study by Shneider and colleagues, published in Pediatrics in 2012, is the key evidence that a standard supplement is not enough. Infants after the Kasai operation received a liquid multivitamin based on TPGS, two millilitres a day, plus vitamin K two point five milligrams three times a week. The researchers then measured how many were insufficient in each vitamin at one, three and six months.

The chart from that study showed that for vitamin A, twenty-nine percent were insufficient at one month, thirty-six percent at three months and thirty-two percent at six months. For vitamin D, insufficiency rose steadily, from twenty-one percent to twenty-seven percent to thirty-seven percent. For vitamin E, it was sixteen percent, nine percent and eighteen percent. For vitamin K, judged by an INR above one point two, it rose from ten percent to seventeen percent to twenty-two percent.

The most important finding concerned the infants who stayed jaundiced. Among those with a total bilirubin above two milligrams per decilitre, only three of twenty-four were sufficient in all four vitamins at three months, and none of twenty-three at six months. So a fixed multivitamin dose is only a starting point. In a jaundiced infant, it is almost never enough on its own.

A companion analysis by Venkat and colleagues in 2014 asked which simple blood test best predicts vitamin deficiency. Total bilirubin did much better than serum bile acids. Its area

under the curve was zero point nine nine eight, against zero point eight two one for bile acids. In practice, a total bilirubin above two milligrams per decilitre is the simplest trigger to intensify supplements and check levels monthly.

Part eleven. Why TPGS matters.

TPGS is vitamin E attached to a water-soluble chain. In water it forms its own micelles, so it does not depend on bile. In 1993, Sokol and colleagues treated sixty cholestatic children whose vitamin E deficiency had not responded to standard oral vitamin E. They gave TPGS at twenty-five international units per kilogram per day. All sixty normalised their vitamin E levels. Over the course of treatment, neurological status improved in twenty-five, stayed stable in twenty-seven and worsened in two, and there were no adverse effects.

TPGS has a second benefit. Because it forms micelles, it can carry other fat-soluble vitamins with it. In 1992, Argao and colleagues showed in eight children that giving TPGS together with vitamin D improved the absorption of vitamin D. So give the other fat-soluble vitamins at the same time as TPGS.

Part twelve. Bone, vitamin K and toxicity.

Bone health deserves its own attention. In a ChiLDReN study by Loomes and colleagues, published in 2019, bone density was lowest in children with chronic intrahepatic cholestasis and in children with Alagille syndrome. After adjusting for body size, the deficit persisted in chronic intrahepatic cholestasis but not in Alagille syndrome. So small size explains part of the problem, but not all of it. In Alagille syndrome, adjusted bone density was lower when bilirubin and bile acids were higher. And children who had had fractures had lower adjusted bone mineral content.

There are also hidden losses beyond the four vitamins. Minerals such as zinc, iron and calcium can follow fat malabsorption. Diuretics used for ascites waste minerals. And cholestyramine, sometimes used for pruritus, binds fat-soluble vitamins in the gut. Review the drug list whenever vitamin levels refuse to correct.

Part thirteen. New drugs: ileal bile acid transporter inhibitors.

A newer class of drugs blocks the ileal bile acid transporter, the protein that reabsorbs bile acids at the end of the small intestine. These drugs are called IBAT inhibitors. They lower serum bile acids and relieve pruritus in progressive familial intrahepatic cholestasis, often shortened to PFIC, and in Alagille syndrome.

For odeixibat, the PEDFIC one trial, published by Thompson and colleagues in 2022, randomised sixty-two children with PFIC. Pruritus improved in fifty-five percent on the drug against thirty percent on placebo, and a bile acid response occurred in thirty-three percent

against none on placebo. An open-label extension called PEDFIC two reported in an abstract that height z-score improved from minus one point six to minus zero point five after forty-eight weeks, but this has not been fully published in a peer-reviewed journal.

For maralixibat, the ICONIC trial in Alagille syndrome, published by Gonzales and colleagues in 2021, showed a fall in serum bile acids of ninety-six micromoles per litre by week forty-eight and a fall in the pruritus score of one point six points. Growth gains have been reported only in a company poster.

Here is the nutritional catch. By blocking the reabsorption of bile acids, these drugs shrink the pool of bile acids available for the next meal. In theory, this can worsen fat and vitamin absorption. Both drug labels require fat-soluble vitamin levels at baseline and during treatment. The odevixibat label reports fat-soluble vitamin deficiency in sixteen percent of children with PFIC on the higher dose, against five percent on placebo. The maralixibat label reports deficiency in ten percent of children with Alagille syndrome over forty-eight weeks. So check vitamins A, D and E and the INR before you start an IBAT inhibitor, and keep checking.

Part fourteen. What the guidelines say.

How do the main documents compare?

On who is covered, there is a material difference. The NASPGHAN and ESPGHAN paper and the AASLD transplant guidelines cover children. The EASL nutrition guideline covers adults with cirrhosis only, so it gives no direct paediatric guidance.

On energy, the documents differ, but mostly because they describe different populations. The paediatric paper sets one hundred and thirty percent of the requirement for age. The AASLD 2014 guideline describes a need for twenty to eighty percent more calories. EASL recommends at least thirty-five kilocalories per kilogram of actual body weight per day for adults.

On protein, they are aligned in the most important respect. None of them recommends restricting protein. The paediatric numbers are one hundred and thirty to one hundred and fifty percent of the requirement, or two to four grams per kilogram per day. The adult number is one point two to one point five grams per kilogram per day.

On anthropometry, they differ. Paediatric documents rely on serial arm circumference and skinfold. The adult guideline prefers a skeletal muscle index measured on computed tomography whenever a scan exists.

On MCT, there is a gap. Only the paediatric paper gives detailed advice. EASL does not address it.

On vitamin D, the difference is material. The paediatric paper gives doses of two to five thousand international units a day, but its printed target has a unit inconsistency. EASL

supplements below twenty nanograms per millilitre and aims above thirty.

On tube feeding and parenteral nutrition, all documents are aligned. Feed by tube when oral intake is inadequate, and use parenteral nutrition when enteral feeding fails.

And on aggressive nutritional support before transplant, the AASLD 2014 guideline gave a strong recommendation, and the 2026 guideline repeats that children with cholestatic disease should receive aggressive nutritional support before transplant to improve outcomes.

Part fifteen. Pearls, and what to do differently in clinic.

Let us go back to the infant from the opening scene, six weeks after the Kasai operation, with a bilirubin of four milligrams per decilitre and thin arms. Here is what this chapter suggests you do.

First, measure the arm. Record mid-upper arm circumference and triceps skinfold at every visit, and plot length carefully. Treat weight as a trend that may be inflated by organs and fluid. Remember that in the Brazilian cohort, skinfold found depletion in sixty-four percent when weight found thirty-three percent.

Second, feed for a hypermetabolic child. Start at about one hundred and thirty percent of the energy requirement for age, and adjust to growth of the arm and of length. Do not restrict protein.

Third, choose fat deliberately. Begin with about thirty percent of fat as MCT, never more than eighty percent, and make sure linoleic acid provides at least three percent of energy. If a rash, poor growth or refractory vitamin deficiency appears, measure the essential fatty acids directly.

Fourth, move early to tube feeding. If energy goals are not met for two weeks, or arm measurements are falling over two to four weeks, start nasogastric feeds, ideally overnight.

Fifth, prove the vitamins with laboratory tests. Use a TPGS-based preparation and give the other fat-soluble vitamins at the same time. Read vitamin E against total lipids, judge vitamin A by the retinol to retinol-binding protein ratio, and give vitamin K even when the INR is normal. Check levels every three to six months, and every month while the bilirubin is above two milligrams per decilitre, because in ChiLDReN almost no jaundiced infant was sufficient in all four vitamins on a standard supplement.

Sixth, treat a falling weight z-score after the Kasai operation as a prognostic sign. In the DeRusso study, infants who went on to transplant or death had a weight z-score of about minus two by six months, against about minus one in those who kept their liver. That is the moment to talk to the transplant team.

Seventh, if the child is listed, measure growth precisely, because the refitted PELD score now counts each step of growth failure, and give aggressive support so that the child reaches the operation with as much muscle as possible.

And eighth, before you start an IBAT inhibitor, check the vitamins, and keep checking them while the child is on it.

Key points.

Children with cholestasis are malnourished for four reasons at once: bile acids do not reach the gut, energy expenditure is about a quarter to a third higher than normal, intake falls, and fuel is diverted from muscle.

Medium-chain fat bypasses the bile defect, but it contains no essential fatty acids, so long-chain fat must still be given.

Weight overestimates nutritional status. Arm circumference and triceps skinfold are the measurements to trust.

Poor growth after the Kasai operation predicts transplant or death, and growth failure counts in transplant allocation.

A standard multivitamin is not enough for a jaundiced infant. Vitamin levels must be measured and doses adjusted.

Questions to test yourself.

Question one. By how much is energy expenditure increased in infants with biliary atresia and in young children with end-stage liver disease, and what energy target does the 2019 position paper recommend? Answer: about twenty-nine percent in the study by Pierro and twenty-seven percent per kilogram in the study by Greer. The target is one hundred and thirty percent of the requirement for age.

Question two. Why is MCT absorbed without bile, and why can it not be the only fat? Answer: medium-chain fats are soluble enough to be absorbed without micelles and go directly into the portal vein. But MCT contains no essential fatty acids, so long-chain fat must still provide at least three percent of energy as linoleic acid.

Question three. In the Brazilian cohort of ninety-one cholestatic children, how did the prevalence of malnutrition differ between weight-for-age and triceps skinfold? Answer: thirty-three percent by weight-for-age against sixty-four percent by triceps skinfold.

Question four. In the DeRusso study, what were the weight z-scores six months after the Kasai operation in infants who kept their native liver and in those with a poor outcome? Answer: minus one point two against minus two point one.

Question five. What is the correct test for vitamin E deficiency in a cholestatic child, and why? Answer: the ratio of vitamin E to total lipids, because high blood lipids can make a deficient child's serum vitamin E look normal. A ratio below zero point six milligrams per gram in a child aged one to twelve years indicates deficiency.

Question six. In the ChiLDReN study of vitamin supplementation after the Kasai operation, how many infants with a total bilirubin above two milligrams per decilitre were sufficient in all four fat-soluble vitamins at three and at six months? Answer: three of twenty-four at three months and none of twenty-three at six months.

Question seven. Why can a child with a normal INR still be vitamin K deficient? Answer: the INR shows that clotting factors are being made, but deficiency relevant to bone can occur in about thirty-nine percent of children despite a normal INR. PIVKA-II is a more sensitive marker.

Question eight. What should you check before starting an IBAT inhibitor, and why? Answer: vitamins A, D and E and the INR, because these drugs reduce the bile acid pool and can worsen fat-soluble vitamin absorption. Both labels require baseline and ongoing monitoring.

End of chapter.